Aashutosh A V

Atlanta, Georgia
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Education

Georgia Institute of Technology **Aug. 2025 – May 2027**
Master of Science in Computer Science (Specialization: Machine Learning) *GPA: 4.0/4.0*

Birla Institute of Technology & Science, Pilani **Aug. 2021 – Jun. 2025**
Bachelor of Engineering in Computer Science (Minor: Data Science) *GPA: 9.1/10*

Experience

Microsoft Research **Jan. 2025 – July 2025**
Research Intern (Advisor: Dr. Ayush Choure) *Bangalore, India*

- Co-invented **ORBIT**, a vector-based online bin-packing algorithm for shard placement; demonstrated a **35% footprint reduction** across **100,000+ Azure VM nodes**, projecting **$100M+ annual savings** for Cosmos DB.
- Co-engineered **LUNA**, a production time-series forecasting pipeline using LightGBM quantile regression and Whittaker-Eilers smoothing to predict peak resource loads under asymmetric loss constraints.
- Contributed to formulating **DRV** (Distressed Resource Volume), a stochastic QoS metric leveraging Kernel Density Estimation and probabilistic bootstrapping to model tail-error rates against real-time workload intensities.
- Built and open-sourced **LOADSTAR**, a high-fidelity policy simulation framework using non-parametric statistical models to benchmark placement algorithms on production traces, reducing compute variance ($\ln(P80/P20) \to 1.34$).

C21U, Georgia Tech **Jan. 2026 – Present**
Graduate Research Assistant (Advisors: Dr. Jeonghyun Lee, Dr. Rod Valente) *Atlanta, GA*

- Building LLM tutoring agents for Socratic Mind, Georgia Tech's production AI tutoring platform, with tool-calling loops, retrieval and context packaging, and scaffolding for state and error recovery.
- Designed a RAG few-shot LLM-as-judge pipeline that compares open-source and API models for ICAP cognitive-engagement classification across logged student–AI tutoring dialogues.
- Built a multi-rater evaluation framework on AWS EC2 with DynamoDB to log and annotate live interaction traces; iterated judges against human annotators until **Krippendorff's** α and Cohen's κ reached **0.7+**.

Hays Lab, Georgia Tech **Jan. 2026 – Present**
Graduate Researcher (Advisor: Dr. James Hays) *Atlanta, GA*

- Building **NVS-Bench** to evaluate sparse-view Novel View Synthesis with diffusion models, studying how context-view selection affects accuracy and multi-view consistency.
- Training **uncertainty predictors** on diffusion models and developing **retrieval algorithms** for context-view selection, outperforming geometric coverage on held-out scenes.

Publications & Patents

Measuring Active Cognition Through Uptake in AI-Mediated Socratic Dialogue | *AERA Annual Meeting* **2027**
- Evaluated cognitive engagement in AI-mediated Socratic dialogue using the ICAP framework and RAG-assisted coding; demonstrated that Interactive AI uptake better differentiates dynamic, exchange-level reasoning than behavioral volume alone.

SCLARO: A Dataset for Grounded Scenario-Level Scene Understanding **2027**
- Proposed a relation-aware VLM with disentangled encoders and EMA-based intra-modal knowledge distillation for compositional scene understanding; curated the Action-Genome Dataset (1.8M image-text pairs).

Analyzing and Optimizing NoSQL Workloads for Cosmos DB | *PVLDB Vol. 19 (Core A*)* **2026**
- Characterized NoSQL workload dynamics at datacenter scale, deriving analytical models for tail-error prediction and replica placement optimization across 100,000+ node Azure Cosmos DB clusters.

Narrating For You: Audio-visual Narrating Face Generation | *WACV 2026 (Core A)* **2026**
- Introduced a multi-entangled latent space that jointly encodes text prompts, voice profiles, and facial identity to enable synchronized joint audio-video generation of talking faces from a single static image.

Workload Distribution Based on Projected Error Counts | *U.S. Patent & Trademark Office (Microsoft)* **2025**
- Novel error-projection scheme that preemptively redistributes NoSQL workloads based on **forecasted error counts**, preventing QoS violations before they occur across distributed database clusters.

Latent Flow Diffusion for Deepfake Video Generation | *CVPR 2024 (Workshop)* **2024**
- Proposed LFD2G, a three-stage deepfake generation pipeline combining latent optical flow prediction, dual Vision Transformer encoding, and flow-conditioned diffusion to disentangle spatial identity from temporal dynamics.

Technical Skills & Achievements

Languages: Python, C/C++, SQL, Bash.
AI/ML Frameworks: PyTorch, JAX, Hugging Face, OpenCV, LightGBM, Scikit-Learn, MLflow, W&B.
Systems & Tools: Amazon AWS (EC2, DynamoDB), Azure, Distributed Systems, Git, HPC.